



Research Diary

Cross-Modal Emotion Analysis in Art and Music

Lab: Computational Intelligence Lab (CILab)

Project: CMEAN

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Introduction

”Is it possible to associate an image that best describes a given song?” This fundamental question lies at the heart of our investigation. Human beings often form rich visual association, music—abstract scenes, emotions, or even specific imagery that resonate with the sound. In this work, we explore whether an artificial system can replicate this process: The goal of the task is to explore approaches to pairing audio and images, focusing on image-based perspectives.

To address this problem, we take advantage of recent advances in multimodal artificial intelligence. Specifically, we use Qwen2-Audio-7B-Instruct, a powerful audio understanding model, to generate textual descriptions from raw audio signals. These captions serve a specific purpose: they act as prompts for a text-to-image diffusion model, guiding the generation of synthetic images that visually reflect the audio content.

To evaluate the relevance of the generated images, we adopt a retrieval-based approach: using CLIP-based embeddings, we compute the semantic similarity between each generated image and a curated set of real artworks. The artwork that is most similar in the shared vision-language embedding space is identified as the best candidate to represent the given audio sample.

In addition to generating images from audio captions, we also explore whether raw audio representations, particularly spectrograms, can provide meaningful insights about the associated artworks. Spectrograms visualize the distribution of energy over time and frequency in the audio signal. Our goal is to explore whether patterns in the spectrogram, such as peaks of energy, correspond visually or semantically to features in the artwork. This analysis helps us to understand if there is a direct relationship between the dynamics of audio signals and the visual elements present in the artwork that best represent the music.

Our results provide insights not only into technical feasibility but also into the deeper multimodal connections that AI can reveal, which may align with how humans perceive and create.

Related Work

Multimodal Audio-Visual Representation Learning

Recent advancements in multimodal learning have focused on aligning audio and visual data within a shared embedding space, enabling cross-modal reasoning [7, 2]. Audio-visual correspondence models typically use contrastive learning to determine whether an audio sample matches a visual input.

WAV2CLIP [7] is a self-supervised model trained to map audio signals into CLIP’s vision-language embedding space, enabling direct cross-modal retrieval between audio and images. This approach allows audio data to be compared with visual representations without requiring explicit paired training data. Similarly, AUDIOCLIP [2] extends the CLIP framework to incorporate audio as a third modality alongside text and images, making it possible to achieve joint embeddings across all three domains.

Although most prior studies explore real-world audio-visual associations, such as linking environmental sounds to corresponding images, our work investigates a more abstract connection: whether an artificial intelligence system can retrieve or associate an artwork that best represents the aesthetic or emotional characteristics of a piece of music. We build on this foundation by proposing a generative retrieval pipeline incorporating **audio-to-text captioning, text-to-

image synthesis, and CLIP-based similarity evaluations** to bridge the gap between musical expression and visual representation.

Qwen-Audio for Audio-to-Text Captioning

Qwen-Audio [1] is a Multimodal Large Language Model (MLLM) specialized in understanding and describing audio content beyond basic transcription. Instead of merely converting sound into text, it captures rich, high-level features from the audio, and uses them to generate captions that capture key musical attributes such as instrumentation, genre, rhythm, and emotional character. This deeper audio understanding enables Qwen-Audio to generate descriptive prompts suitable for downstream tasks. For instance, we used it to generate captions from audio files, which were later used to generate representative images.

Text-to-Image Diffusion Models in Multimodal Generation

Diffusion models such as Stable Diffusion and DALL·E have revolutionized text-to-image generation by iteratively refining noise towards a target prompt [8]. In this work, Stable Diffusion serves as the generative backbone, transforming Qwen-Audio’s textual descriptions into images that visually embody the input music. This approach ensures that generated images reflect abstract musical qualities rather than direct sonic patterns.

CLIP for Cross-Modal Understanding

CLIP [5] (Contrastive Language–Image Pretraining) is a foundational model trained to align visual and textual information within a unified embedding space . In our study, CLIP was utilized to evaluate the similarity between generated images and real artwork by extracting features from the images and attempting to form meaningful pairs . This retrieval-based methodology strengthens the connection between AI-generated visual content and established artistic works.

Spectrograms as Visual Representations of Audio

Spectrograms encode time-frequency energy distributions as 2D images and are widely used in music analysis and speech recognition [3]. Although typically used for classification tasks, recent research has explored their potential for multimodal learning. In our work, we analyze whether spectrograms contain structural elements that align with artistic compositions, investigating whether spectral peaks correspond to visually salient regions in artwork [6].

Datasets

The Free Music Archive (FMA) dataset is a large-scale collection designed for Music Information Retrieval tasks. It consists of 106,574 tracks, spanning 161 genres. The dataset provides full-length, high-quality audio, along with precomputed features and metadata. We used the dataset in the Small version. Particular attention was given to the "genre_top" label assigned to each track, ensuring consistency and handling missing values to maintain data integrity.

ArtGraph is an artistic knowledge graph designed to integrate structured information from WikiArt and DBpedia, offering a rich dataset for art analysis, retrieval, and classification. It

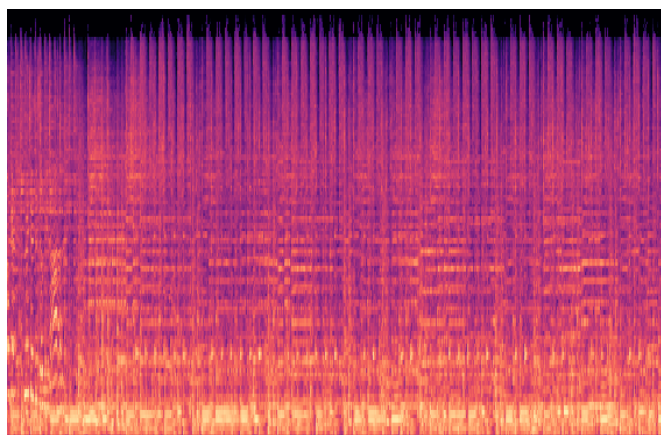


Figure 1: Example of a mel-spectrogram generated from an audio track in the FMA dataset.

consists of 116,475 artworks across 18 genres and 32 artistic styles, making it one of the most comprehensive resources available for computational art research. Beyond digitized images, ArtGraph provides detailed metadata, including artist names, historical context, stylistic attributes, and painting schools.

We applied a stratified sampling approach to both datasets, selecting 100 instances from ArtGraph and 100 instances from FMA for an initial analysis. This process was carried out as an exploratory phase, but it can be extended to the entire dataset for broader evaluation.

Data Preprocessing

Spectrogram and Caption Generation

Each track in the FMA dataset was converted into a mel-spectrogram (example in Figure 1), a representation that maps sound frequencies onto a perceptually meaningful scale. This offers a structured and visually interpretable form of audio. The conversion process employed the Librosa library, which enabled the extraction of spectrograms subsequently saved as image files. These images were then used in experiments alongside digitized artworks and CLIP, allowing for the exploration of potential correspondences between auditory and visual elements.

Furthermore, we generate textual descriptions of each audio clip, focusing on the general mood, emotional tone, and the instruments used in the track. We use **Qwen2-Audio-7B-Instruct**, an instruction-tuned variant of Qwen2-Audio, capable of producing natural and interpretable captions directly from raw audio inputs.

Each audio clip, provided in its original .mp3 format, was processed using the torchaudio library to ensure consistency across the dataset. Files were converted to mono and resampled at 16kHz. The raw audio files were then passed directly to Qwen2-Audio-7B-Instruct, which handled the processing and caption generation autonomously.

We set the maximum number of tokens to 128 and use the following prompt to guide the model in generating detailed and relevant captions in a chat-style format:

System: You are a helpful assistant that describes audio clips in a simple and descriptive way, focusing on general mood, atmosphere, and broad characteristics rather than technical details.

User: [AUDIO] + Generate the caption of this audio along with its instruments and its mood (emotions).

An example of a caption generated by the model is shown below:

"A mellow instrumental track featuring acoustic guitar and soft ambient pads. The mood is calm and introspective."

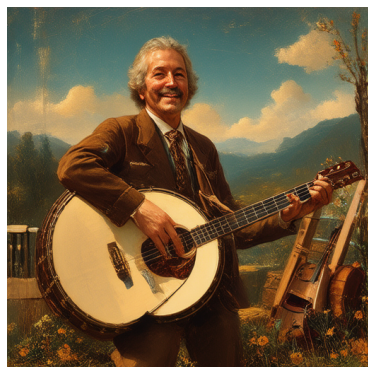
Image Generation from Audio Captions

Once the audio captions were generated, the next step was to “visualize” them by generating corresponding images. For this purpose, we used the **Stable Diffusion 3.5 Medium** model, which provides improved coherence and prompt alignment over earlier versions.

We used the `stabilityai/stable-diffusion-3.5-medium` as a diffusion model for generating an image representative of the audio description.

The model was used without additional prompt engineering to preserve the original semantic meaning of the audio descriptions. This results in using the raw caption obtained from Qwen2-7B-Instruct as a prompt for the Diffusion Model.

Images were generated in a loop over the caption dataset (100 entries), with each image saved and indexed by its corresponding audio file ID. Generation parameters included 15 steps, a guidance scale of 4.5, and a resolution of 512×512 pixels. Examples of generated images are illustrated in Figure 2.



Caption for audio file 105718.mp3:
"The audio features a lively folk tune with acoustic guitar and banjo in E major key and 4/4 time signature, conveying a happy and nostalgic mood. Instruments include mandolin, fiddle, and harmonica as well."



Caption for audio file 063250.mp3:
"The audio features a psychedelic rock track with punk and folk influences, set in G major with a tempo of around 89 BPM. The chord progression is complex and varied, contributing to an overall dreamy atmosphere. Instruments include drums, bass, electric guitar, and acoustic guitar."



Caption for audio file 002099.mp3:
"The audio features a complex instrumental blend including acoustic guitar, tabla, sitar, and violin, set in Ab major with a tempo of 84.5 BPM, and a chord progression of G# major, D# minor, C# major, and A minor. The overall mood is described as happy, love, summer, and tropical."

Figure 2: Examples of images generated by Stable Diffusion 3.5 based on audio captions from Qwen2-Audio-Instruct.

Embeddings Extraction and Similarity Computation

To evaluate the semantic alignment between the generated images and the existing digitized artworks, we use CLIP to extract embeddings from both.

This process involves computing embeddings for all digitized artworks in the selected subset of ArtGraph digitized artworks, as well as for all images generated from audio captions. The embeddings are then stored to avoid redundant computations during iterative experiments.

These embeddings are later used to compute pairwise cosine similarity scores, allowing us to identify the closest matches based on semantic similarity.

Evaluation Metrics

To complement semantic similarity, we introduce a *visual coherence metric* that measures pixel-level similarity between the generated image I_g and its closest digitized artwork I_a . The *Visual Coherence Score* quantifies how structurally similar the two images are by calculating the normalized Mean Squared Error (MSE) between their grayscale versions. This score highlights how well the shapes and textures (luminance patterns) in the original digitized artwork are preserved in the generated image, adding to the semantic comparison from CLIP embeddings.

Both images are first resized, converted to grayscale, and normalized to the $[0, 1]$ range. The grayscale conversion is performed using the function `rgb2gray`, which applies weighted coefficients to the RGB channels to approximate human perception of brightness:

$$\text{rgb2gray}(\text{img}) = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$

The grayscale conversion formula `rgb2gray` was firstly used by the NTSC (National Television Standards Committee) [4]. This transformation ensures that the luminance distribution is preserved, making the similarity metric robust to color variations.

Let I_g and I_a be the resized grayscale images normalized in the range $[0, 1]$:

$$I_g = \frac{\text{rgb2gray}(\text{gen_resized})}{255.0}, \quad I_a = \frac{\text{rgb2gray}(\text{art_resized})}{255.0}$$

The Mean Squared Error (MSE) between the two grayscale images is then computed as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_a[i] - I_g[i])^2$$

where N is the number of pixels.

Finally, the *Visual Coherence Score* S_{visual} is defined by normalizing the MSE with respect to a maximum threshold MSE_{max} :

$$S_{\text{visual}} = \max \left(0, 1 - \frac{\text{MSE}}{\text{MSE}_{\text{max}}} \right)$$

where MSE_{max} is an empirically chosen parameter that sets the maximum tolerated dissimilarity. In this work, we set it to 0.5.

The score has these characteristics:

- $S_{\text{visual}} = 1$ indicates perfect visual coherence (zero pixel difference);
- $S_{\text{visual}} = 0$ indicates complete visual dissimilarity;
- Intermediate values represent varying levels of visual similarity, inversely related to pixel-wise error.

By converting images to grayscale and normalizing them, this metric enables a uniform and meaningful comparison across image pairs. The *Visual Coherence Score* serves as a complementary quantitative measure alongside CLIP-based semantic similarity, providing further validation of the relationship between generated images and real artworks.

Comparison of Generated Images, Spectrograms, and Digitized Artworks

In this phase of the analysis, we aimed to evaluate how well the visual representations—both generated and derived directly from audio—align with digitized artworks in the ArtGraph dataset. To this end, we designed three distinct types of cross-modal comparisons, each probing a different hypothesis regarding the relationship between audio and visual content:

1. **Generated Image vs Digitized Artwork:** We used Stable Diffusion to generate images from audio captions, which were automatically extracted using a pre-trained audio captioning model. The objective was to assess whether images generated from textual descriptions of audio carried visual characteristics similar to digitized artworks. Each generated image was compared to all digitized artworks using CLIP embeddings, and the most semantically similar digitized artwork was selected.
2. **Spectrogram vs Digitized Artwork:** In this second comparison, we directly used audio spectrograms and compared them with digitized artworks. This tested whether a direct visual representation of sound could semantically align with visual art without relying on intermediate transformations. Comparisons were made in the CLIP embedding space using cosine similarity.
3. **Spectrogram vs Recolored Digitized Artwork:** To improve the visual alignment, we recolored the digitized artworks using the Inferno colormap—the same one used to generate the spectrograms. This perceptual transformation enhanced visual contrast and aimed to emphasize structural features that could match better with the spectrograms.

For each comparison, we computed semantic similarity using CLIP embeddings and retrieved the top 5 most similar digitized artworks. To maintain assignment diversity, each digitized artwork was used at most once. In addition to semantic similarity, we computed the previously introduced *Visual Coherence Score*, which offers a complementary low-level evaluation by capturing structural similarities not reflected in embedding-based metrics.

By combining these three analyses, we obtained a more comprehensive understanding of how audio-based visual representations—both generated and direct—relate to artistic styles, and how perceptual transformations (like colormap adjustments) can enhance cross-modal alignment.

Results are reported in Table 1. Overall, generated images (Gen_Img-Artw.) achieve the highest semantic similarity (mean emb_sim = 0.498), suggesting that text-guided image synthesis

Table 1: Summary of similarity and visual coherence metrics across the three evaluation setups. Gen_Img stands for Generated Image. Artw. stands for Artwork. Spect stand for Spectrogram. Rec_Artw stands for recoloured artwork. Std.Dev stands for Standard Deviation. emb_sim stands for Embedding Similarity, vis_score stands for visual score. vis_mse stands for visual mse.

Analysis	Metric	Std. Dev.	Mean	Min	Max
Gen_Img-Artw.	emb_sim	0.098200	0.498200	0.240800	0.749800
	vis_score	0.216179	0.649900	0.000000	0.929200
	vis_mse	0.111715	0.176100	0.035400	0.604700
Spect.-Artw.	emb_sim	0.077065	0.442600	0.259100	0.668800
	vis_score	0.205718	0.691500	0.000000	0.953300
	vis_mse	0.103041	0.154300	0.023400	0.505300
Spect.-Rec_Artw.	emb_sim	0.077065	0.442600	0.259100	0.668800
	vis_score	0.158907	0.759000	0.145000	0.943000
	vis_mse	0.079453	0.120500	0.028500	0.427500

captures meaningful high-level concepts from the audio. This setting indicates a high standard deviation on the visual_score, likely due to the variability with which the diffusion model generates the image based on the obtained caption.

Spectrogram-to-artwork alignment (Spect.-Artw.) exhibits lower semantic similarity compared to the generated images. This is because spectrograms are energy-based representations, capturing the peaks and distributions of audio energy rather than conveying any intrinsic semantic meaning. As a result, their alignment with digitized artworks is purely structural rather than conceptually driven, unlike the images generated from captions.

The third analysis is about shifting the digitized artwork in the same color space of the spectrogram. This doesn't denote changes in embeddings similarity, but improves something in the visual coherence score side. This is probably due to the fact the having the same color map produces a better visual coherence in terms of structural properties and chromatic aspects.

Standard deviations reveal substantial variability across samples, especially for visual scores, highlighting the heterogeneous nature of visual mappings across different audio inputs.

Clustering and Cross-Modal Analysis of Genres and Artistic Styles

To investigate the relationship between musical genres and artwork styles, we performed a clustering analysis based on the metadata and similarity scores obtained from our pipeline.

We combined three data sources:

- The dataset with audio IDs and their closest digitized artworks found by visual similarity;
- The metadata for audio tracks from the FMA dataset;
- The metadata for digitized artworks from the ArtGraph dataset, including artwork file-names and styles.

This merging linked each audio sample to its genre and to the style of its most visually similar digitized artwork.

Proportional Distribution of Art Styles Across Music Genres

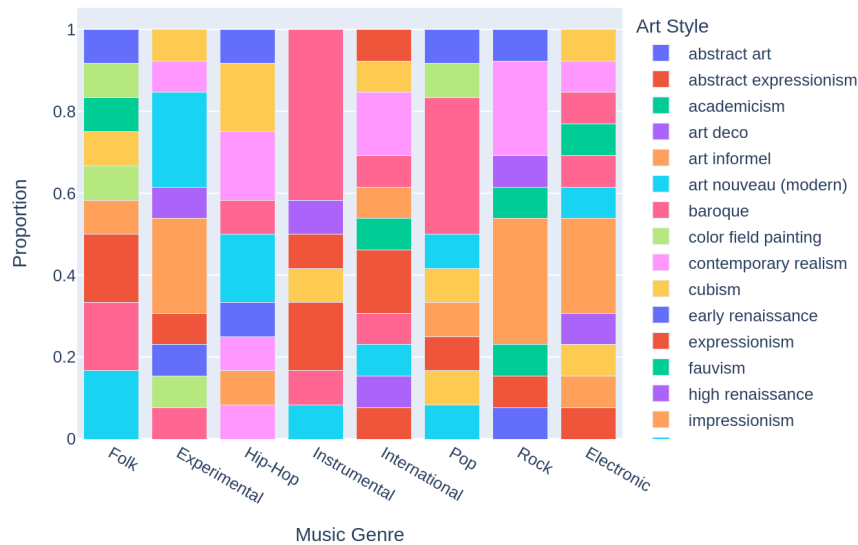


Figure 3: Distribution of artistic styles across different musical genres.

Using the merged dataset, we computed the frequency distribution of digitized artwork styles associated with each musical genre. This was achieved by grouping the data by genre and digitized artwork style and counting the occurrences. To better interpret the associations, we converted the counts into percentage distributions normalized per genre. This allows for clearer insight into which artwork styles predominantly correspond to certain musical genres.

We visualized the resulting distributions as a stacked bar chart (Figure 3). The colors and labeled values show the percentage of each style within each genre. The bar chart highlights patterns where particular genres are linked to specific visual styles, indicating meaningful connections between audio features and artistic visuals. For instance, some music genres like Rock, Electronic and Experimental often relate to abstract or impressionist art, while others like Pop and Instrumental are more associated with realistic or surreal styles.

Discussion

Several interesting observations emerge from analysing the results obtained. We provide insights on different cases of audio-digitized artwork pairing quality and the factors influencing similarity scores.

Generated Image-Digitized Artwork

For the audio file *011789.mp3* (Figure 4), you can say that it is actually an audio containing hip-hop, jazz and soul music as described in the caption. This results in a generated image that represents a kind of pseudo musical score, which is associated with a score of 0.552 to the digitized artwork *kuzma-petrov-vokdin-violin-1918.jpg* which is a digitized artwork representing a violin. The score in this case is a modest value that in reality could have been much lower since the digitized artwork found does not really reflect the semantic content of the audio.

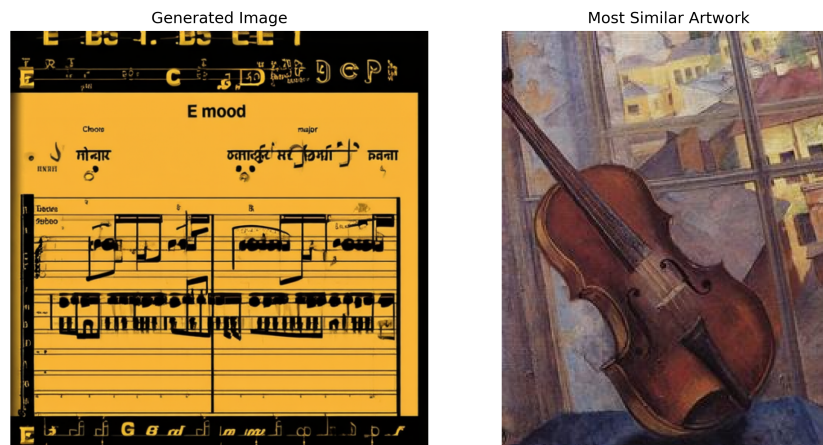


Figure 4: Audio Caption: The audio features a blend of hip-hop, jazz, and soul music set in E minor with a tempo of 95.2 BPM. The chord progression includes E major, C major, D major, B minor, A major, and G major. Instruments include bass, drums, guitar, piano, saxophone, and trumpet. The mood is described as cool.

For the audio file *071252.mp3* (Figure 5), you can say that it is actually a rap/hip-hop track that leaves you an happy and party mood. In this case the most similar digitized artwork, *eric-fischl_barbecue.jpg*, represents very satisfactorily the mental image that the audio leaves you when you listen to it, in particular the digitized artwork is describing a situation of happiness and chilling (a barbecue in a pool) like in the song. The similarity value, 0.44, does not fully reflect the goodness of the digitized artwork identification. While the visual coherence is more in line with the real audio-image association, having a value of about 0.79.

For the audio file the audio *113336.mp3* (Figure 6), we can see that it contains electronic music, also very abstract. This results in a generated image that is also very abstract, which is associated with an unexpectedly high similarity value (0.74) to the digitized artwork *olivier-debre_esquisse-boulogne-1988.jpg*. Both the digitized artwork and the generated image are abstract, but on a level of meaning they denote no particular evidence. This also results in a not particularly high visual coherence value.

We can notice that in the three examples, the one with the shortest prompt (Figure 5) is the

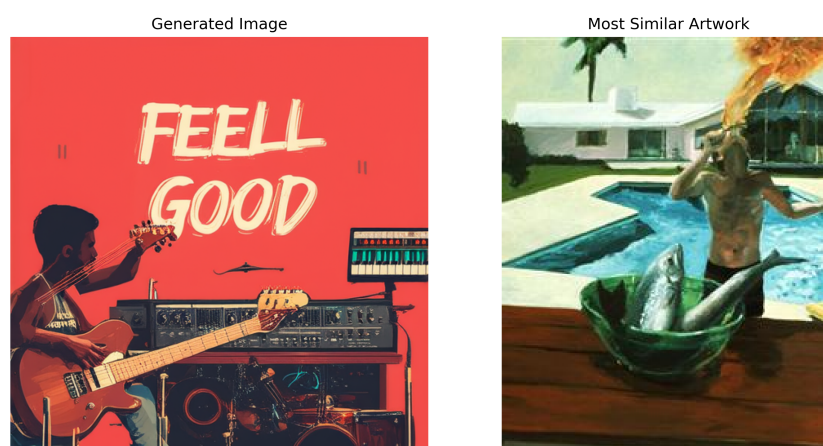


Figure 5: Audio Caption: The audio is a hip-hop track with a feel-good mood, featuring instruments such as an acoustic guitar, bass, drum machine, piano, synthesizer, and violin.

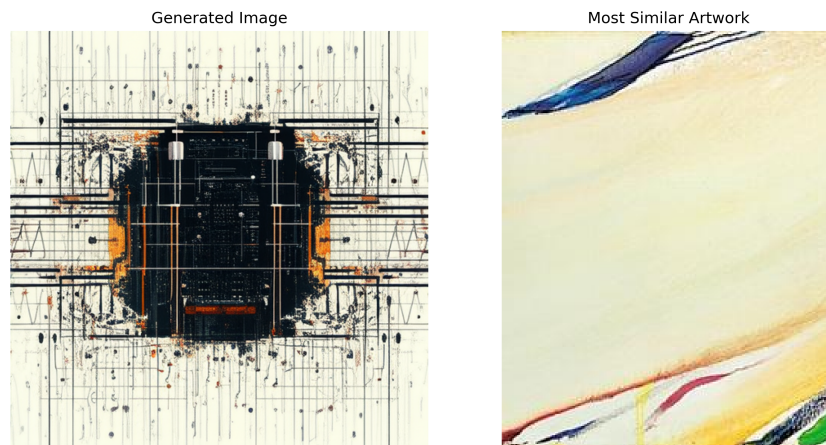


Figure 6: Audio Caption: The audio is an experimental electronic track with abstract and glitch-hop elements in Db major at a tempo of 82 BPM with a 4/4 time signature. It features a complex chord progression including A minor, C major, and B minor.

one with the most appropriate pairing of audio and digitized artworks. This may suggest that it is probably best to focus on a few but essential aspects of audio. In addition, we can say that in some specific cases the digitized artwork reflects very much the style of the music, but for purposes of matching with digitized artwork this could result in lower values of similarities.

Spectrogram-Artwork

Subsequently, we explore the results from the comparison between the spectrograms and the digitized artworks.

For the audio file *011789.mp3* (Figure 7), the result is that the similarity between the spectrogram and the most similar digitized artwork is 0.66. But we know that this result it's not a really demonstration of semantic link between spectrogram and digitized artwork. Probably it is caused by the total abstraction that is present naturally on the spectrogram but also, in this case, on the digitized artwork. The result to consider more in this case is the visual score, that is 0.85. This is probably due to the colors in the digitized artwork, that are very similar to the colors in the spectrogram.

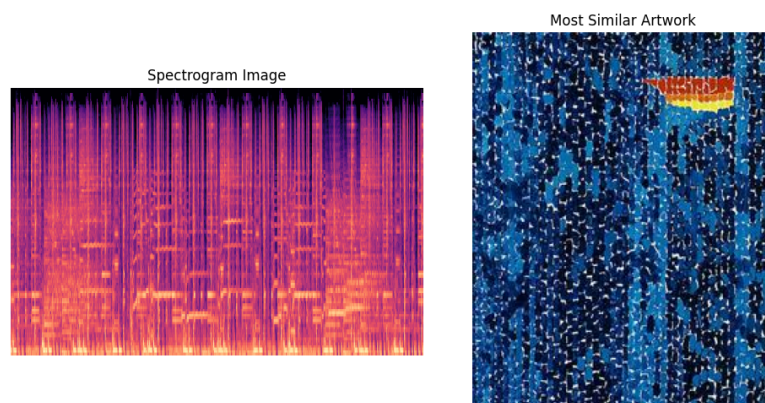


Figure 7: The spectrogram of the audio 011789.mp3 and the most similar artwork by similarity embedding.

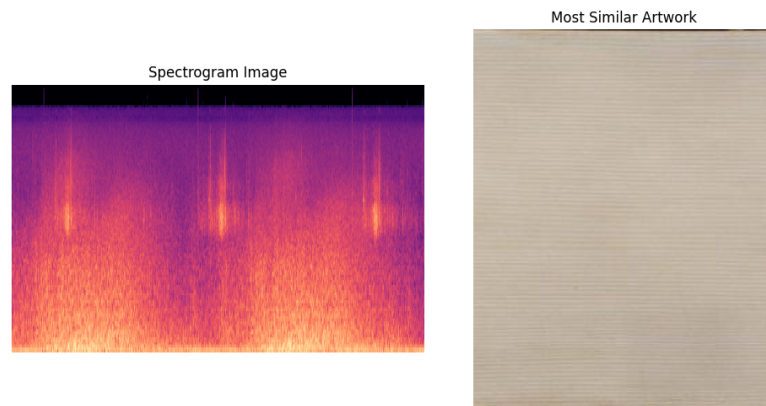


Figure 8: The spectrogram of the audio 0711252.mp3 and the most similar artwork by similarity embedding.

For the audio file *0711252.mp3* (Figure 8), This case is similar to the first example in terms of similarity score, that is 0.59. The difference from the first example is in the complexity of the spectrogram, that is greatly less than the first spectrogram. This results in an digitized artwork without any meaning and with only a color. The result on the visual score tell us that this type of score on this setting depends greatly on the colors in the digitized artwork.

For the audio file *113336.mp3* (Figure 9), the spectrogram is surely more complex than the first and also the second spectrogram. The similarity score between this spectrogram and the selected digitized artwork is 0.58, but this is, like in the other cases, not linkable with a semantic assumption. The visual score is 0.73, more than the second spectrogram, and this is due to the major presence of colors of the spectrogram in the artwork.

In general, we can say that the spectrogram himself doesn't tell us something relevant about some semantic linking with a digitized artwork. The results on the coherence score depends mostly on the colors in the digitized artwork.

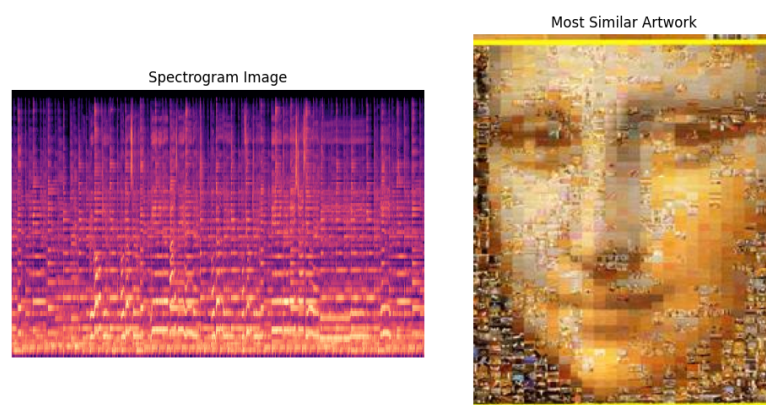


Figure 9: The spectrogram of the audio 011789.mp3 and the most similar artwork by similarity embedding.

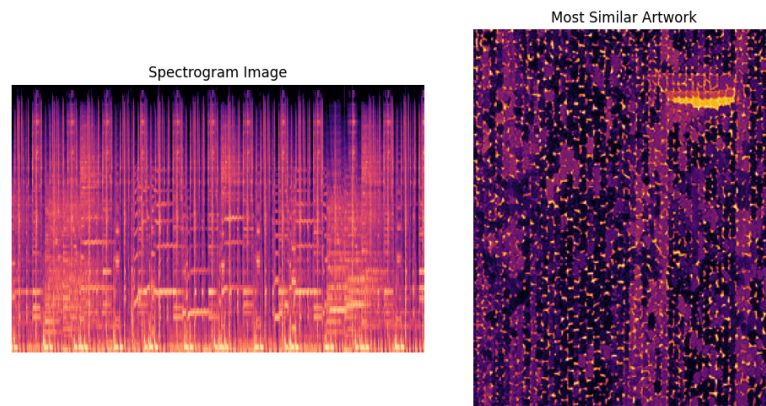


Figure 10: The spectrogram of the audio 011789.mp3 and the most similar artwork changed to the color map "inferno"

Spectrogram-Recoloured Digitized Artwork

Lastly, in the third dataset, digitized artworks were adjusted to match the spectrogram colormap (*Inferno*) to examine whether this transformation influences similarity metrics. This was done to allow pixels in the artwork to emulate the behavior of pixels in spectrograms. In other words, we aim to replicate the concept of energy in spectrograms: very bright pixels should represent areas of higher energy, while darker pixels correspond to lower energy. In this way, we attempt to map artworks to music based on the energy they "emit."

For the audio file *011789.mp3* (Figure 10), the change of color map doesn't change the visual coherence score, probably because the matching of colors was already quite marked. The values of similarity remain the same.

For the audio file *071252.mp3* (Figure 11), the value of visual coherence score decreases to 0.51. The reason of this is probably the monochromy of the digitized artwork. The values of similarity remains the same.

For the audio file *113336.mp3* (Figure 12), the value of the visual coherence score is amplified by the change of the color map. The reason of this can be the fact that there were already enough colors of the colormap. The values of similarity remains the same.

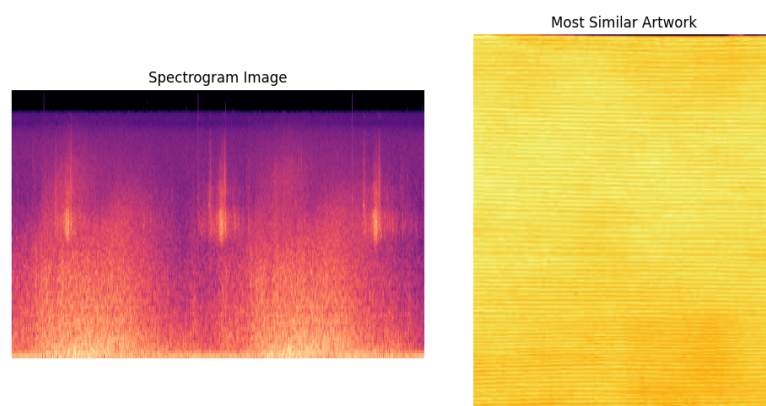


Figure 11: The spectrogram of the audio 071252.mp3 and the most similar artwork changed to the color map "inferno"

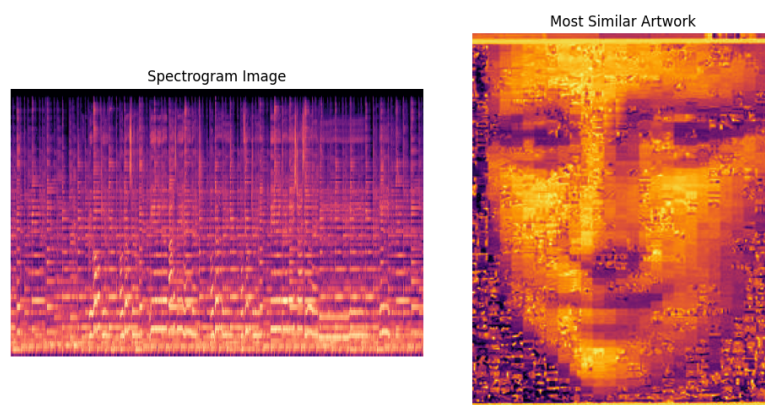


Figure 12: The spectrogram of the audio 113336.mp3 and the most similar artwork changed to the color map "inferno"

The fact that in all the cases the values of similarity respect to the digitized artwork doesn't change in affect to the change of color map, is due to the fact that changing the color map doesn't modify the semantic representation of the digitized artwork. These results indicate that modifying the digitized artwork's color map to match that of the spectrograms reinforces the idea that their association is purely based on chromatic and geometric properties. This confirms that spectrogram-based alignment does not incorporate semantic meaning, but rather relies on structural and visual similarities.

Exploring Visual and Musical Style Associations

An additional clustering analysis mapped the relationships between **musical genres** and **artistic styles**, showing distinct preferences:

- **Rock** aligns strongly with *Impressionist* and *Expressionist* movements, likely due to their emphasis on **intense emotional expression, bold contrasts, and dynamic movement**. Much like Rock music's raw energy, these artistic styles favor *gestural brushstrokes, distorted forms, and dramatic color shifts*, mirroring the genre's expressive nature and rebellious undertones.
- **Electronic** and **Experimental** music favor *abstract styles*, particularly *Color Field painting* and *Neo-Expressionism*. The *non-representational and layered compositions* found in these artistic movements resemble Electronic music's complex soundscapes, synthesized textures, and evolving tonal progressions. These styles often rely on **gradual shifts in intensity, large chromatic fields, and structural abstraction**, echoing the genre's focus on atmospheres rather than direct narrative.
- **Instrumental genres** correspond with *Realist* styles, reflecting their **precision, structure, and clarity**. Realism's emphasis on meticulously detailed representation resonates with the **harmonic balance, technical accuracy, and composed arrangements** found in instrumental music. Much like classical compositions and orchestral pieces, Realist artworks maintain *clear subject matter, defined spatial relationships, and controlled lighting*, reinforcing a sense of order and intentionality.
- **Narrative-driven genres** (Hip-Hop, Folk, Jazz) display a preference for *Symbolism* and *Surrealism*, indicating an affinity for **metaphorical storytelling, unconventional**

perspectives, and layered meanings. Hip-Hop’s lyrical depth, Folk’s cultural narratives, and Jazz’s improvisational storytelling parallel **the symbolic representations and dreamlike imagery** often explored in these artistic movements. The abstraction of Surrealism and the allegorical nature of Symbolism complement music that conveys *social messages, emotional reflection, and personal expression through intricate symbolism.*

Conclusion

This work investigates whether an artificial system can effectively map digitized artworks to music by integrating multimodal techniques. Through a pipeline incorporating audio-to-text captioning, text-to-image synthesis, and cross-modal similarity retrieval, we explored how musical elements align with existing artistic styles and representations.

The analysis reveals several key findings regarding the relationship between musical genres and artistic styles. Rather than generating arbitrary visuals, the system consistently maps musical characteristics to well-defined artistic movements, demonstrating a strong preference for compositional structure, contrast, and abstraction in its mappings. Cluster analysis highlights systematic genre-style affinities, showing that Rock aligns with Impressionism and Expressionism, likely due to their dynamic and emotionally charged aesthetics, while Electronic and Experimental favor abstract styles such as Color Field painting and Neo-Expressionism, which reflect the layered and atmospheric nature of these genres. Additionally, Instrumental music exhibits a strong correspondence with Realism, emphasizing structured composition and detail, whereas narrative-driven genres like Hip-Hop, Folk, and Jazz tend to align with Symbolism and Surrealism, which emphasize metaphor and storytelling.

Finally, an investigation into spectrogram-based matching indicates that while spectrograms provide a structured visual representation of audio, their similarity to digitized artworks is largely determined by geometric and chromatic properties rather than deep conceptual meaning. These findings reinforce the idea that structural and perceptual elements influence musical-visual relationships more than explicit semantic content.

While this study has demonstrated structured relationships between musical genres and artistic styles, several avenues remain open for further exploration:

- *Expanding Cross-Modal Embedding Spaces:* Future research could refine the multimodal mapping between music and visual art by incorporating deeper semantic embeddings that account for thematic and emotional interpretations rather than structural intensity alone.
- *Fine-Tuning Generative Models for Music-to-Image Translation:* Instead of relying solely on text-based captions, new approaches could explore direct music-to-image generation using specialized diffusion models trained on curated datasets that blend audio features with artistic imagery.
- *Investigating Human Perceptual Alignment:* Conducting user studies on how humans perceive AI-generated images in relation to music could help refine similarity metrics by incorporating subjective interpretations alongside computational assessments.
- *Analyzing Temporal-Visual Associations:* Since music unfolds over time, future research could explore whether temporal structures in audio, such as rhythm and dynamics, correspond with visual composition techniques like movement, perspective, or focal balance in artworks.

- *Exploring Genre-Specific Image Characteristics:* A more granular study could analyze whether subgenres within broad musical categories (e.g., psychedelic rock vs. punk rock, ambient electronic vs. techno) exhibit distinct visual correlations, refining the framework for multimodal synthesis.

These directions aim to enhance the depth and accuracy of multimodal mappings between sound and visual representation, contributing to a richer understanding of cross-modal creativity.

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